PARKTOWN BOYS' HIGH SCHOOL

Department of Mathematics

Cycle Test

28 May 2025

Subject	Mathematics
Grade	12
Topic	Calculus
Marks	50
Time	1 hour

Examiner	Mrs S Rowell
Moderator	Dr G Nixon Mrs M Trollope

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 3 questions and 4 pages. Please ensure that your question paper is complete.

2. Answer ALL the questions.

3. Number the answers correctly according to the numbering system used in this question paper.

4. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.

5. Answers only will NOT necessarily be awarded full marks.

6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.

7. If necessary, round off answers to TWO decimal places, unless stated otherwise.

8. Diagrams are NOT necessarily drawn to scale.

9. Use the mark allocation to guide the length of your solution.

10. Write neatly and legibly.

11. Rule off after each question.

QUESTION 1

1.1 If $f(x)=3x^2-2x,$ determine $f'(x)$ from first principles. (5)

1.2 Determine:

1.2.1 $\frac{d}{dx}\left[(3x-4)^2\right]$ (3)

1.2.2 $g'(x)$ if $g(x)=8\sqrt{x}-\frac{2}{x^3}$ (3)

1.2.3 $\frac{ds}{dt}$ if $s=-3t^4+pt-q$ (2)

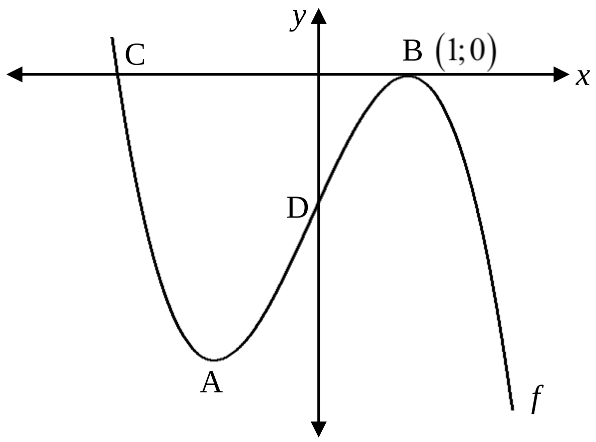
1.3 Determine the equation of the tangent to $h(x)=x^3-5x^2+3x-1$ at $x=1.$ (4)

1.4 Sketch the graph of $y=(x+1)^3+8,$ showing all relevant points. (4)

[21]

QUESTION 2

The graph of $f(x)=ax^3+bx-4$ is sketched below. A and B $(1;0)$ are stationary points of *f*. B and C are x – intercepts and D is the y – intercept of *f*.



2.1 Write down the coordinates of D. (1)

2.2 Show that $a=-2$ and $b=6.$ (4)

2.3 Determine the coordinates of A. (4)

2.4 Determine the coordinates of C. (3)

2.5 Show that D is the point of inflection of *f*. (2)

2.6 For which value(s) of x is:

2.6.1 f decreasing? (2)

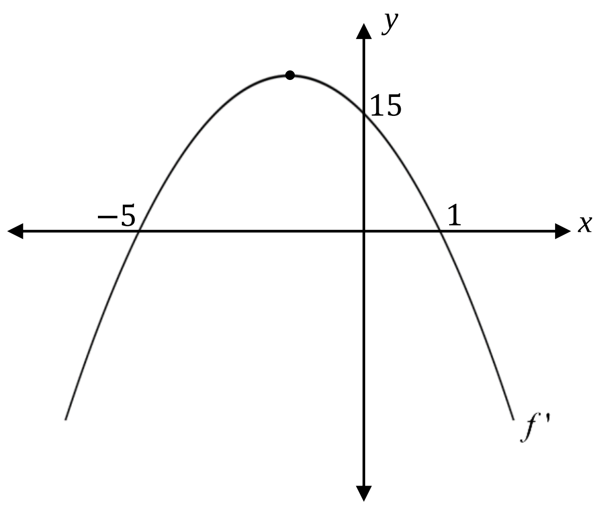
2.6.2 $x \cdot f'(x) > 0$ (2)

2.7 For which value(s) of k will $f(x) = k$ have two distinct roots? (2)

[20]

QUESTION 3

The graph of $y = f'(x)$ is sketched below. The graph intersects the y – axis at 15 and the x – axis at -5 and 1.



3.1 What is the gradient of the tangent to f at $x = 0$? (1)

3.2 At which value of x does f have a point of inflection? (2)

3.3 If f passes through the origin, draw a rough sketch of f. (4)

3.4 For which values of x is f concave up? (2)

[9]